TERTIARY INFOTECH ACADEMY

LAB 10
Operate DNS and Content-Delivery Failover

AWS Certified SysOps Administrator Associate Training
TGS-2024051413 · Version v5.0

Scenario
Northstar Retail needs a controlled failover path for a public web endpoint and must distinguish DNS, origin and cache effects during an incident.

Objective
Apply K2 and A2 by configuring health-aware DNS and CloudFront behaviour, then proving client-visible routing and cache outcomes.

Deliverable
A Route 53 health-check definition, failover record plan, CloudFront operations checklist and measured failover report.

Services and tools
Route 53 · Health Checks · CloudFront · S3 · CloudWatch

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Review `DNS-FAILOVER-PLAN.md`; record hosted zone, primary/secondary endpoints, TTL, health criteria and rollback authority.
```
open DNS-FAILOVER-PLAN.md
```

Evidence note: _____

Step 2 — Validate `health-check.json`; replace the trainer-provided test endpoint and unique caller reference.
```
python3 -m json.tool health-check.json
```

Evidence note: _____

Step 3 — Create the health check and confirm CloudWatch health status before associating any record.
```
aws route53 create-health-check --caller-reference SysOps-Lab10-REPLACE_SUFFIX --health-
check-config file://health-check.json | tee evidence/health-check.json
```

Evidence note: _____

Step 4 — Review `failover-records.json`; confirm PRIMARY and SECONDARY identifiers, evaluate-target-health setting and TTL.
```
python3 -m json.tool failover-records.json
```

Evidence note: _____

Step 5 — Apply the trainer-approved learner-zone record change and record the change ID.

```
aws route53 change-resource-record-sets --hosted-zone-id REPLACE_ZONE_ID --change-batch
file://failover-records.json | tee evidence/record-change.json
```

Evidence note: _____

Step 6 — Query the name repeatedly from two resolvers; record answers and TTL before failure injection.

```
for n in {1..5}; do date -u +%FT%TZ; dig +noall +answer REPLACE_NAME; sleep 5; done | tee
evidence/dns-before.txt
```

Evidence note: _____

Step 7 — Run the bounded primary failure approved by the trainer; measure health-check change and client answer transition.

```
for n in {1..20}; do date -u +%FT%TZ; dig +short REPLACE_NAME; sleep 10; done | tee
evidence/dns-failover.txt
```

Evidence note: _____

Step 8 — Inspect CloudFront response headers for cache status, age and request ID; distinguish cached content from origin recovery.

```
curl -I https://REPLACE_DISTRIBUTION/ | tee evidence/cloudfront-headers.txt
```

Evidence note: _____

Step 9 — Complete `FAILOVER-REPORT.md` with observed TTL, health transition, client transition and limitation.

```
open FAILOVER-REPORT.md
```

Evidence note: _____

Step 10 — Capture DNS health and CloudFront evidence as JPEGs; run validation and restore the baseline.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

Health and record evidence are linked, failover timing is measured rather than assumed, cache and DNS effects are distinguished, baseline is restored, and validation emits LAB 10 PASS.

Acceptance criteria

- Uses learner-owned hosted zone
- Primary and secondary identifiers are explicit
- TTL is measured
- Cache response is distinguished from origin
- Rollback restores baseline

Evidence checklist

□ Health check

□ record change

□ DNS before/failover results

□ CloudFront headers

□ completed failover report

□ JPEG evidence

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.